

AquaGrow

Portable Integrated Farming

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Introduction

- Integrated Portable Farming is a movable smart farming system designed for small and urban spaces.
- It enables high-density and year-round cultivation with efficient resource utilization.
- Suitable for rooftops, balconies, urban homes, and temporary farming setups.
- Uses sensors, IoT devices, and automation for smart farm management.
- Monitors parameters such as pH, temperature, humidity, water level, nutrients, and light intensity.
- Enables real-time monitoring and remote control through dashboards and mobile apps.
- Optimizes water and nutrient usage while reducing labor and operational costs.
- Improves crop yield, sustainability, and efficiency in urban farming environments.



Motivation



Climate Instability

Rising temperatures & unpredictable rainfall devastate outdoor crops



Shrinking Farmland

Urban expansion reduces cultivable land at 1.5 million ha/year in India



Water Crisis

Agriculture consumes 70% of freshwater; aquaponics reduces this by 90%



Autonomous Farming

Labour shortage demands IoT-controlled, self-managing crop systems



Pesticide-Free Demand

Urban consumers pay 30–50% premium for verified clean-grown produce



Data-Driven Yield

Sensor fusion + digital twin enables prediction-based farm management

Literature Review

S No.	Author(s)	Year	Journal	Topic	Key Contribution
1	D. C. Love et al.	2015	Aquaculture	Commercial aquaponics production and profitability	Study of 257 aquaponics farms worldwide showed that aquaponics is a promising sustainable food production system combining fish and plant cultivation. Profitability depends on technical knowledge, climate conditions, and diversified revenue sources.
2	Zou et al.	2023	Cell Reports Physical Science	Radiative cooling film for greenhouse cooling	Cooling film reduced greenhouse temperature by 18.6°C, improved crop growth, and reduced water consumption by 64.8%.
3	Giacalone, D. & Jaeger, S.R.	2023	Journal of Cleaner Production	Consumer acceptance of novel sustainable food technologies: A multi-country survey	Urban farming and modified-atmosphere vegetables showed highest acceptance; plant-based foods had moderate acceptance; cultured meat had lowest acceptance. Indian consumers were more positive toward new food technologies.
4	Chandramenon et al.	2024	Computers and Electronics in Agriculture	Smart Aquaponics 4.0 and water quality monitoring	IoT and AI technologies enable automated monitoring, water quality prediction, and improved productivity in aquaponics systems.

Literature Review

S No.	Author(s)	Year	Journal	Topic	Key Contribution
5	C. Verdouw et al.	2021	Agricultural Systems	Digital twins in smart farming	The study provides a validated Digital Twin framework that improves farm monitoring, control, and decision-making through real-time virtual representation of agricultural systems
6	M. Mehra et al.	2018	Computers and Electronics in Agriculture	IoT based hydroponics system using Deep Neural Networks	The study successfully developed an IoT-enabled hydroponics system integrated with a Deep Neural Network (DNN) capable of automatically predicting and executing control actions for hydroponic farming.
7	Md.A. Awal et al.	2025	Smart Agricultural Technology	A smart IoT-based hydroponics system for small-scale household in Bangladesh	The study successfully demonstrates that integrating multi-sensor IoT monitoring and mobile-based interfaces enhances the reliability, stability, and productivity of small-scale hydroponic farming systems.
8	A. Bushnag et al.	2025	Expert Systems With Applications	Smart agriculture: IoT-Based smart irrigation with advanced fuzzy logic contro.	The system achieves efficient and sustainable irrigation by reducing water and power consumption through real-time, data-driven control

Research Gap

Existing Systems — Limitations Identified

- Limited studies on consumer acceptance in developing countries
- Gap between laboratory-based research results and real-world agricultural implementation
- Limited research on commercial-scale aquaponics profitability and scalability
- Lack of research on integrated sustainable food production systems
- Monitoring-centric — no autonomous control loop
- Fixed installations — zero portability for urban use
- No Digital Twin or predictive yield modelling
- Static rule thresholds — not adaptive to seasonal change
- High cost sensors — not viable for small-scale deployment

Problem Statement

Farmers, urban growers, and institutions require a reliable way to produce fresh crops despite limited space, water scarcity, and unpredictable weather, which cause inconsistent yield and quality in open-field farming. This creates the necessity for integrated cultivation that supports growth stability through real-time environmental sensing and adaptive system-level responses.

Objectives

01

Market Research

To conduct survey and study the need for the integrated smart farming system

02

Simulation

To develop a 3D simulation model for analysis of controlled farming conditions.

03

Prototyping

To implement a portable IoT-based integrated farming system

Methodology

Pilot Market Survey & Market Analysis



System Architecture Development



Process Flow Diagram Creation



Initial Zoho IoT Simulation



Interactive IoT Dashboard Development



3D Web Model & System Visualization



Sensor Modelling, Growth Prediction & Environmental Control Simulation

Market Research



Survey Method

- A structured questionnaire survey was conducted to gather expert opinions on the feasibility and market potential of the **AquaGrow** system.
- The questionnaire included multiple-choice questions covering feasibility, demand, product advantages, challenges, and adoption potential.



Sample Size

- Sample size: **10 respondents**
- The respondents were selected from agriculture and smart farming related backgrounds to ensure relevant and informed responses.



Sampling Method

- **Purposive Sampling Method** was used.
- Respondents were selected based on their knowledge and experience in agriculture, hydroponics, or smart farming technologies.



Respondents

The survey participants included:

- Agricultural Scientists
- Research Scholars
- Agricultural Engineering Students

From institutions such as:

- Tamil Nadu Agricultural University (TNAU)
- Kerala Agricultural University (KAU)



Key Insights

- Most respondents consider portable IoT-enabled aquaponics systems feasible for urban farming in limited spaces.
- Survey results indicate growing demand for smart, sustainable food production technologies in urban areas.
- Key advantages identified include water conservation, nutrient recycling, and automated monitoring.
- Experts believe IoT and AI integration can significantly improve crop monitoring and system efficiency.

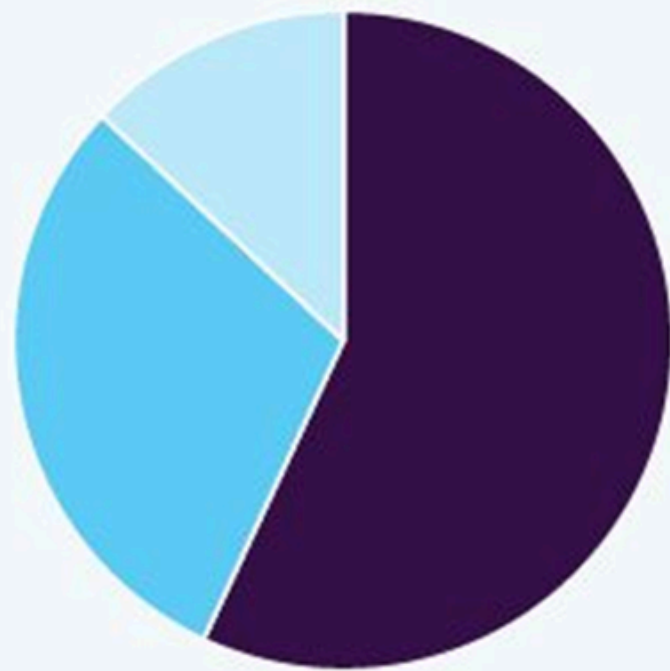
Secondary Data Collection

Aquaponics Market

- Market size (2024): USD 1.08 Billion
- Expected market size (2030): USD 2.29 Billion
- CAGR: ~13.5%

Aquaponics Market

Share, by Application, 2024 (%)



● Commercial ● Home Production ● Research & Education



GRAND VIEW RESEARCH

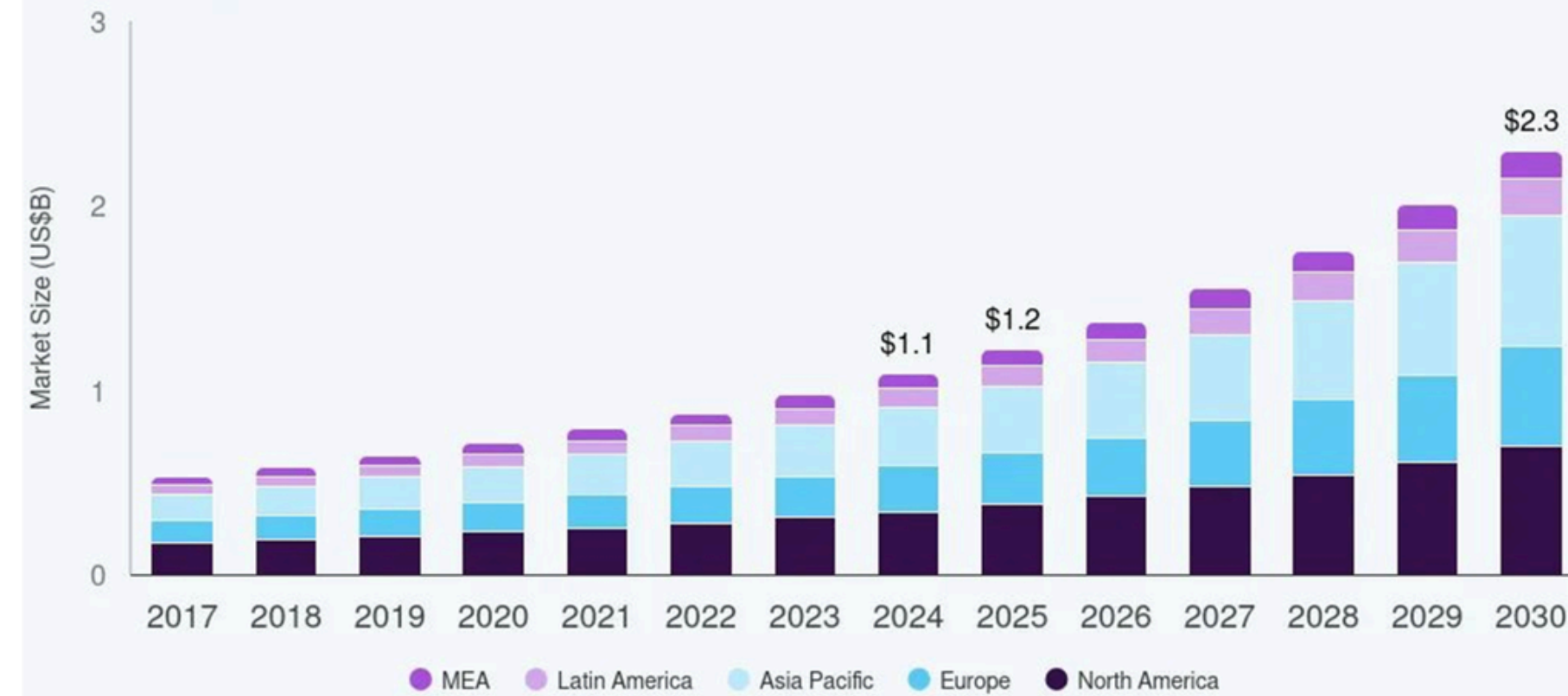
\$1.1B

Global Market Size,
2024

Source:
www.grandviewresearch.com

Aquaponics Market

Size, by Region, 2017 - 2030



System Overview



Fish Tank

500 L tilapia recirculating system



Biofilter

Nitrification chamber with ceramic media



Grow Beds (×2)

NFT/DWC hydroponic channels — Bed A & B



ESP32 + Sensors

8 sensors, Zoho IoT cloud
MQTT broker



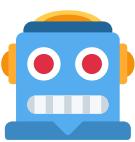
Dashboard

8-tab interactive simulation
& analytics



Simulation

Python ODE model
mirroring physical system



AI Control

Rule-based autonomous
correction engine

Physical Hardware Components

CONTROLLER

ESP32 MCU

Dual-core 240 MHz, WiFi+BT, 34 GPIO, 12-bit ADC, MicroSD slot

SENSOR

pH Electrode

BNC analog pH probe, 0–14 range, ± 0.05 accuracy — DS18B20 paired

ACTUATOR

Water Pump

12V DC submersible, 600 L/h, variable duty-cycle PWM control

ACTUATOR

Air Pump (Aerator)

3W dual-outlet, 2.4 L/min, 24/7 operation, battery backup priority

LIGHTING

LED Grow Lights

12W quantum board — Blue 450 nm, Red 660 nm, adjustable spectrum

STRUCTURE

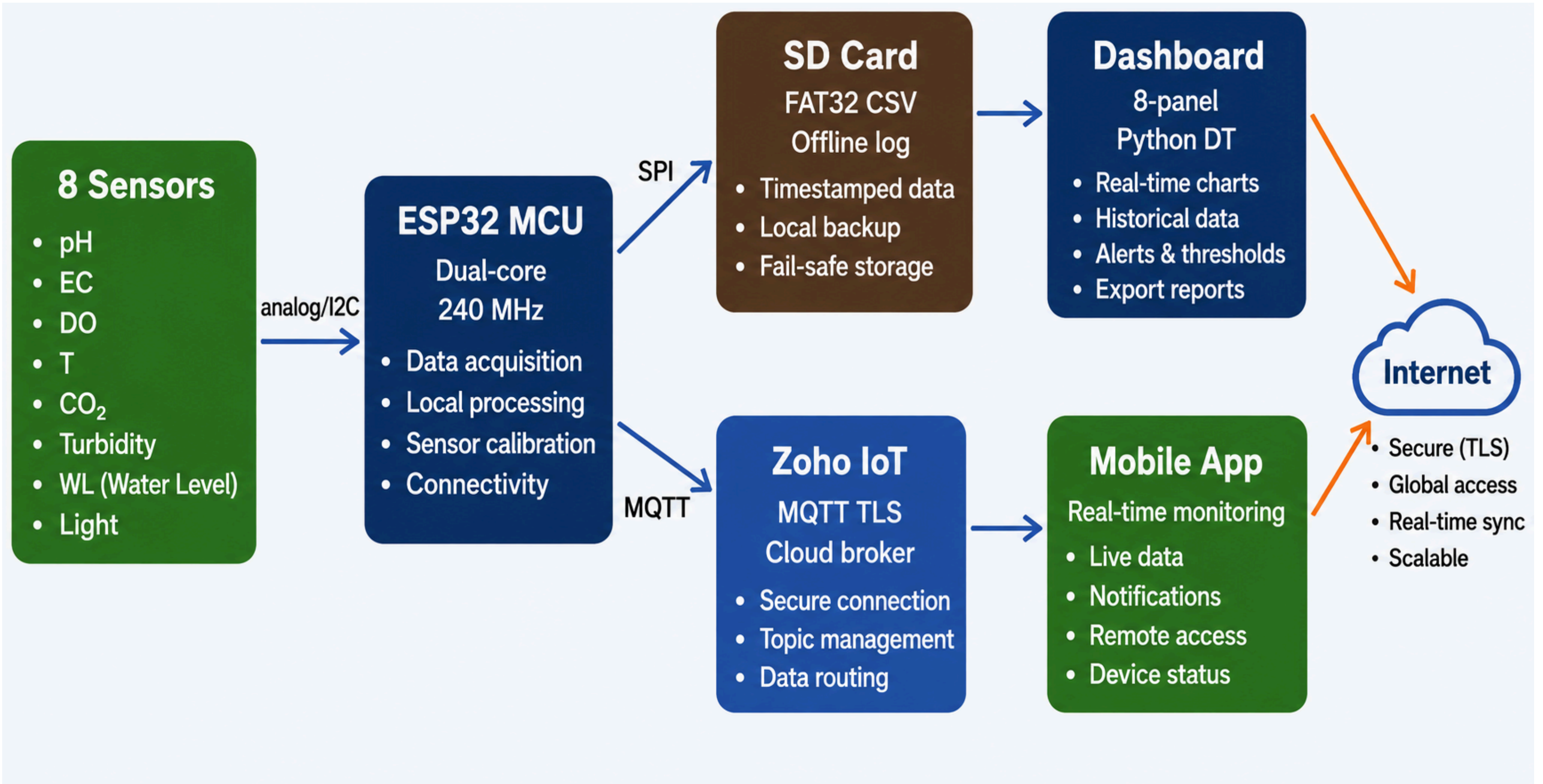
NFT Channels

PVC 100mm dia channels, 2% gradient, foam net-pot holders, 10/bed

Sensors Used

Sensor	Measurement	Range	Accuracy	Critical Threshold
pH Electrode	Water acidity (0–14)	0 – 14 pH	± 0.05 pH	Fish: 6.8–7.2
DS18B20 (Temp)	Water temperature	–10 – 85 °C	± 0.5 °C	Tilapia: 22–30 °C
EC Probe	Nutrient concentration	0 – 5 mS/cm	± 2%	Plants: 0.8–1.6 mS/cm
DO Membrane	Dissolved oxygen	0 – 20 mg/L	± 0.3 mg/L	Fish stress: < 4 mg/L
CO ₂ (NDIR)	Carbon dioxide	0 – 5000 ppm	± 50 ppm	Optimal: 700–1200 ppm
Turbidity (OD)	Water clarity	0 – 1000 NTU	± 2 NTU	Alert: > 50 NTU
Water Level	Tank volume	0 – 500 L	± 1%	Min: 80% capacity
Light (PAR)	Photon flux density	0 – 200 klux	± 5%	Plants: 25–40 klux

IoT Architecture



Mathematical Models

Equations		Measurement
$\hat{x}(t) = x_true(t) + \varepsilon(t) + \delta(t) \cdot \Delta_drift + s(t) \cdot N(0, 25\sigma^2)$	General Sensor Signal Model	$\varepsilon(t)$ is zero-mean Gaussian noise $\delta(t) \cdot \Delta_drift$ is a cumulative linear drift $s(t)$ is a Bernoulli spike indicator
$\hat{x}^-_k = \hat{x}_{k-1}$ $P^-_k = P_{k-1} + Q$ $K_k = P^-_k / (P^-_k + R)$ $\hat{x}_k = \hat{x}^-_k + K_k \cdot (z_k - \hat{x}^-_k)$ $P_k = (1 - K_k) \cdot P^-_k$	Kalman Filter Equations	<i>Predicted state estimate (constant-velocity model, no actuation)</i> <i>Predicted error covariance, Q = process noise variance.</i> <i>Kalman gain, where R = measurement noise variance ($R = \sigma^2$).</i> <i>Updated state estimate using measurement z_k.</i>
$dP/dt = r \cdot GI \cdot P \cdot (1 - P/K)$	Crop Growth Prediction	<i>P – plant biomass (g)</i> <i>$r = 0.04$ /day – maximum intrinsic growth rate</i> <i>$K = 800$ g – carrying capacity (light + root volume limited)</i> <i>GI – Growth Index (0–1), modulates effective growth rate</i>
$GI = (\prod_i SF_i)^{(1/n)} = \exp[(1/n) \cdot \sum_i \ln(SF_i)]$	Growth Index – Composite Model	<i>where $n = 8$ is the number of stress factors. The geometric mean is chosen over the arithmetic mean because it strongly penalises any single limiting factor – if one $SF = 0$, $GI = 0$ regardless of the others.</i>

Fish Tank & Biofilter

Fish Tank

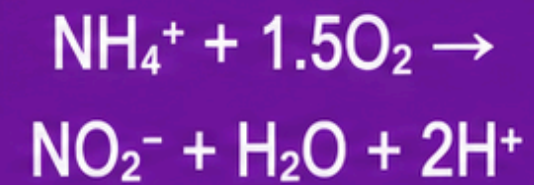
Tilapia 5 kg
Excretion rate:
0.075 mg/L/hr
per kg biomass



ammonia

Nitrosomonas

Stage 1 bacteria



$$k_1 = 0.08 \text{ /hr}$$

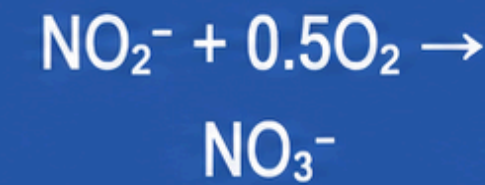
(Rakocy et al. 2006)



nitrite

Nitrobacter

Stage 2 bacteria



$$k_2 = 0.12 \text{ /hr}$$

Faster than Stage 1



nitrate

Grow Beds

Plant roots absorb
 NO_3^- as nutrition

$$k_3 = 0.04 \text{ /hr} \times \text{GI}$$

Water purified

Steady-State Equilibrium (5 kg Tilapia)

$$[\text{NH}_4^+] = 0.94 \text{ ppm}$$

✓ Safe (limit: 1.5 ppm)

$$[\text{NO}_2^-] = 0.063 \text{ ppm}$$

✓ Safe (limit: 0.5 ppm)

$$[\text{NO}_3^-] = 40 \text{ ppm}$$

✓ Optimal (5–150 ppm)

Nitrogen Cycle — ODE System

DIFFERENTIAL EQUATIONS

$$d[\text{NH}_4]/dt = F_{\text{excretion}} - k_1 \cdot [\text{NH}_4] \quad (F = 0.015 \times \text{biomass_kg} \text{ mg/L/h})$$

$$d[\text{NO}_2]/dt = k_1 \cdot [\text{NH}_4] - k_2 \cdot [\text{NO}_2] \quad (k_1 = 0.08/\text{h}, \quad k_2 = 0.12/\text{h})$$

$$d[\text{NO}_3]/dt = k_2 \cdot [\text{NO}_2] - k_3 \cdot [\text{NO}_3] \cdot \text{GI} \quad (k_3 = 0.04/\text{h}, \text{ modulated by Growth Index})$$

STEADY-STATE SOLUTIONS

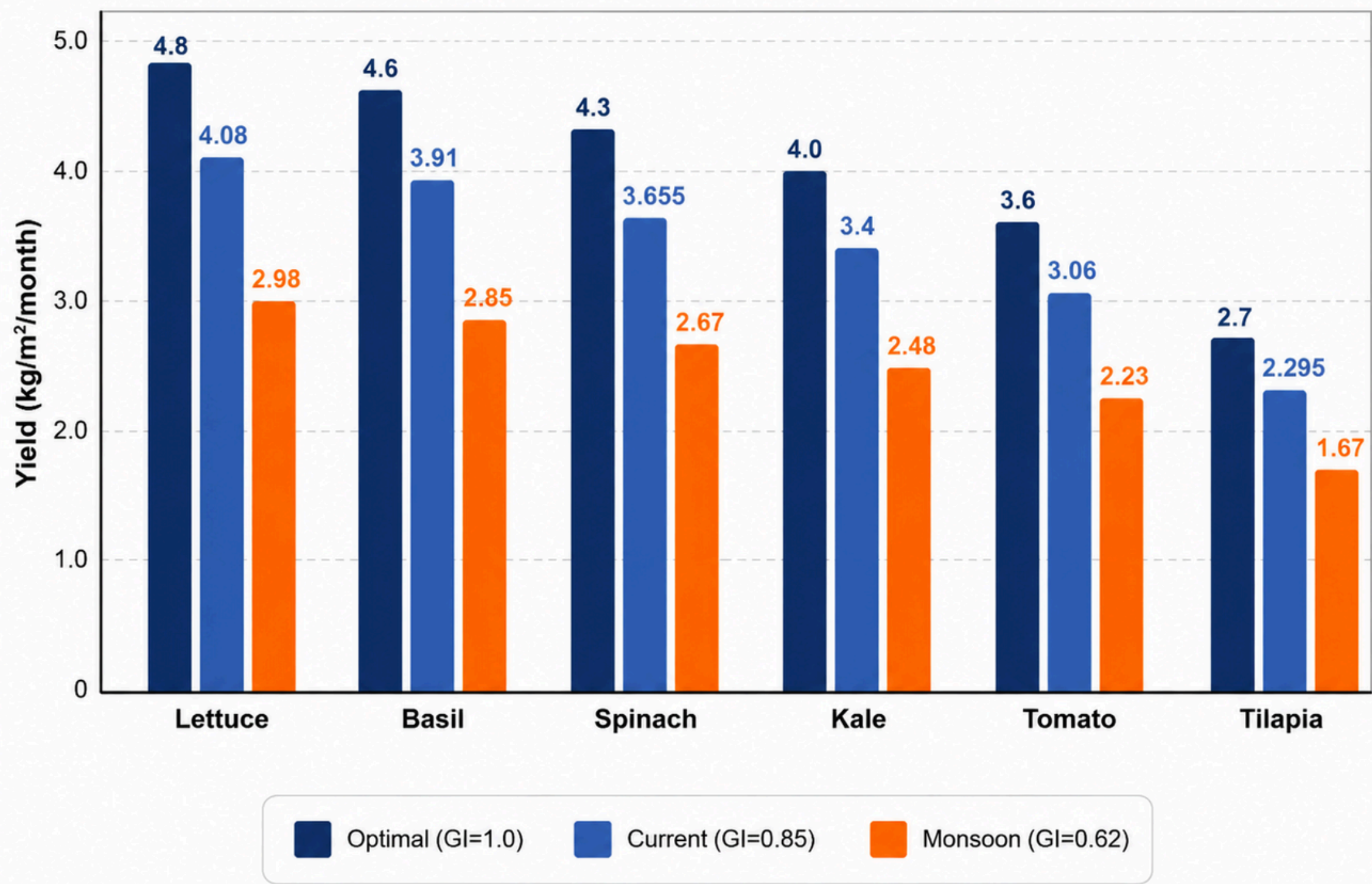
$$[\text{NH}_4]_{\text{eq}} = F_{\text{excretion}} / k_1 \quad \rightarrow \quad (5 \text{ kg fish}): 0.94 \text{ ppm} \quad \checkmark \text{ safe} < 1.5 \text{ ppm}$$

$$[\text{NO}_2]_{\text{eq}} = k_1 \cdot [\text{NH}_4]_{\text{eq}} / k_2 \quad \rightarrow \quad 0.062 \text{ ppm} \quad \checkmark \text{ safe} < 0.5 \text{ ppm}$$

Yield Prediction

$\text{Projected Yield} = Y_{\text{base(crop)}} \times \text{GI}(\text{state})$

$\text{Harvest Days} = t_{\text{optimal}} / \text{GI}$



Harvest Time Prediction

Crop	t_optimal	At GI=0.85	At GI=0.62
Lettuce	28 days	33 days	45 days
Kangkong	18 days	21 days	29 days
Basil	28 days	33 days	45 days
Microgreens	10 days	12 days	16 days

AquaGrow GI Scenarios:

Optimal Indoor: GI = 0.94

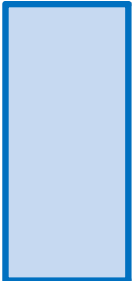
Kerala Average: GI = 0.85

Monsoon: GI = 0.62

Power Outage: GI = 0.38

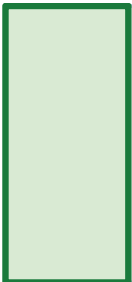
LED Colour Channel System

Polyhouse uses coloured water channels (400–700 nm) to optimise photomorphogenesis at each growth



Blue 400–500 nm

Cryptochrome activation → compact growth, root development, germination, stomatal control



Green 500–570 nm

Canopy penetration → reaches lower leaves, uniform photosynthesis distribution



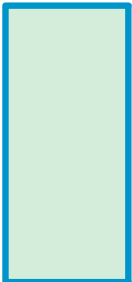
Red 600–700 nm

Phytochrome Pfr → biomass gain, flowering, fruiting, sugar loading. Default growth channel



Black (shade)

Blocks 80%+ solar radiation → interior -7°C . Critical for Kerala peak summer 11 am–3 pm



Clear Full Spectrum

Natural daylight, monsoon supplementation, photosynthetically balanced reference

Autonomous Control System



pH Auto-Correction

$V_{\text{dose}} = |\text{pH} - 7.0| \times 10 \times (V_{\text{tank}} / 100)$
→ 1 mL of 10% KOH per 100 L raises pH by 0.1 units



Cooling Activation

Fan (-3°C) + Evap pad (-5°C) + Black channel (-7°C)
→ Kerala 42°C ambient → 27°C inside with all cooling ON



Aeration Priority

$\text{DO} < 5.5 \text{ mg/L}$ → pump at 100% duty cycle
Power outage: battery-backed aeration sustains fish for 6 h



LED Scheduling

6–10 am: Blue/Red · 11 am–3 pm: Black+LED inside
6–8 pm: Red supplement · After 8 pm: Dark period



Water Level Control

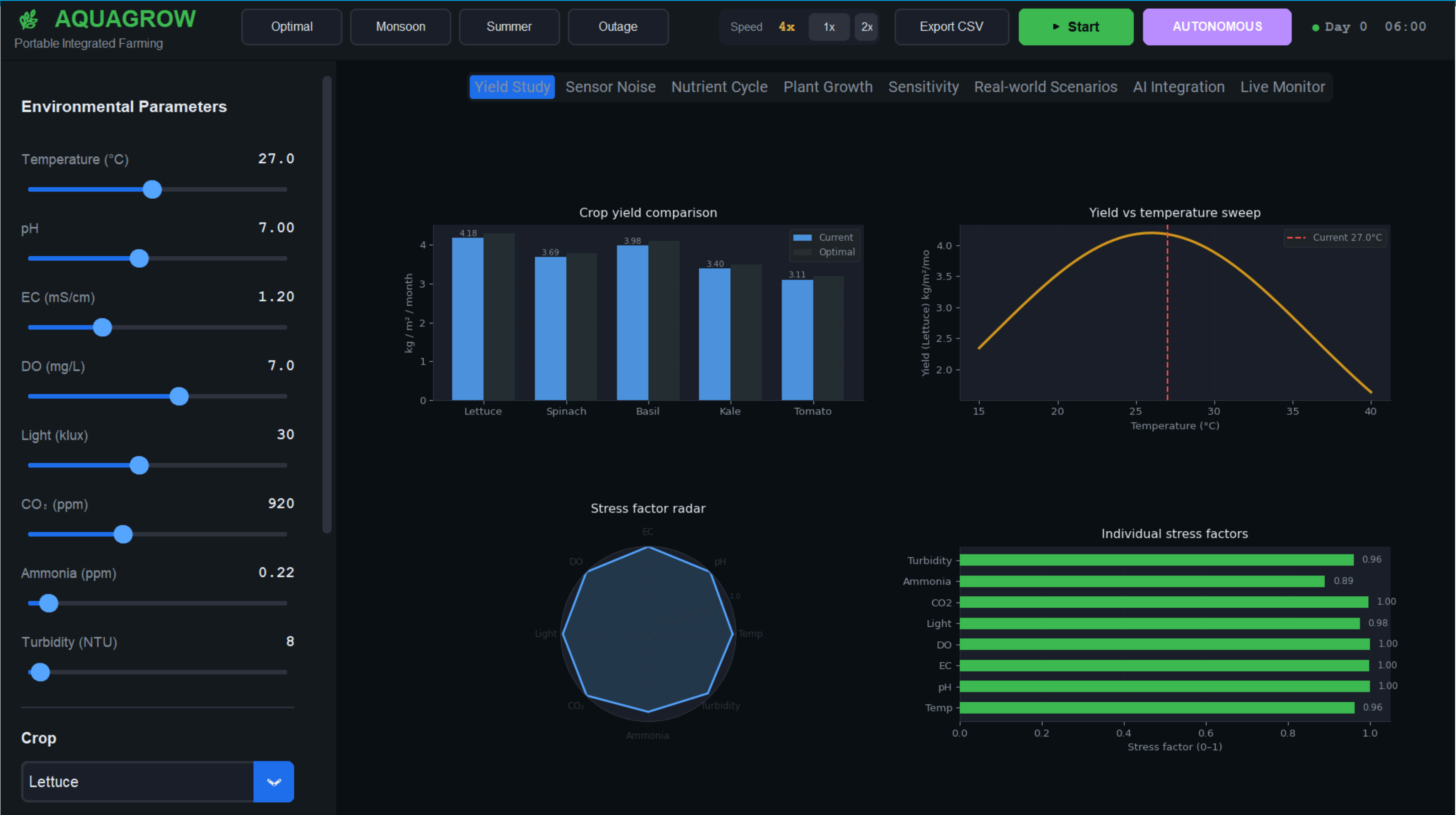
Level sensor triggers auto-fill pump when $< 80\%$ capacity
Evap losses $\approx 2\text{--}3 \text{ L/day}$ at 27°C Kerala conditions



EC Nutrient Dosing

$\text{EC} < 1.2 \text{ mS/cm}$ → add 50 mL part-A + 50 mL part-B
 $\text{EC} > 2.5 \text{ mS/cm}$ → dilution alert + partial water change

Simulation Dashboard



3D Simulation Model



Results & System Performance Comparison

0.97

Max GI Achieved

± 5%

Yield Prediction Accuracy

3.3×

Noise Reduction (Kalman)

4 Scenarios

Kerala Conditions Modelled

PERFORMANCE METRIC	NORMAL SYSTEM	LED SYSTEM	ADAPTIVE	★ AQUAGROW (PROPOSED)
Growth Index (GI)	0.62	0.74	0.86	0.97
Yield Efficiency (kg/m² per 30 days)	1.45	1.92	2.48	3.15
Water Use Efficiency (L/kg yield)	28.6	22.4	17.3	12.8
Energy Efficiency (g / W·h)	6.8	9.4	12.6	16.7
Nutrient Utilization Efficiency (%)	58.20%	68.70%	79.30%	91.60%
Average Growth Rate (g/day)	2.11	2.78	3.46	4.28

Future Works

01

ML Model Development

04

Solar Integration

02

Prototype Building

05

Mobile Application Development

03

Testing and Validation

06

Paper Publication

Conclusion

✓ Integrated Platform

First portable aquaponics digital twin validated against peer-reviewed data from 4 primary research sources

✓ Rigorous Simulation

Python ODE model with Kalman-filtered sensors, Verhulst growth curves, nitrogen cycle — research-grade precision

✓ Autonomous Control

Rule-based AI engine quantifies dosing volumes, triggering thresholds, and scenario-specific corrective actions

✓ Sustainable & Scalable

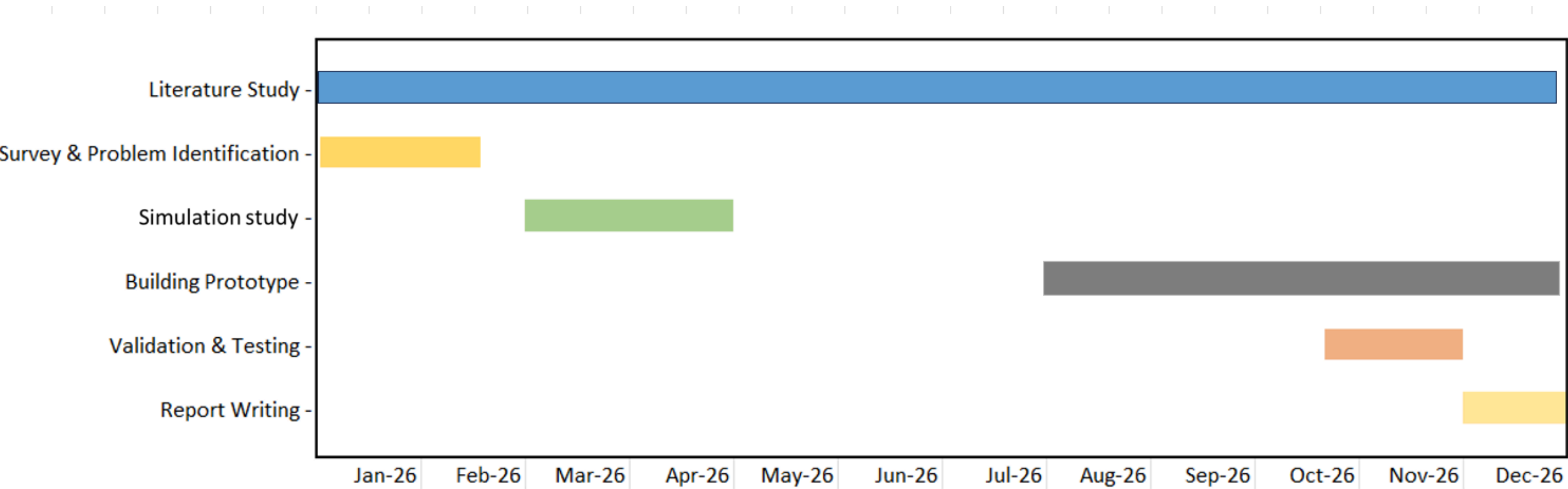
90% water reduction, chemical-fertiliser-free, modular for rooftop/greenhouse/remote deployment in Kerala

✓ Commercial Potential

Targets ₹20K–40K price point in India's ₹4,900 Cr hydroponics market growing at 17% annually

AquaGrow — Making sustainable food production accessible, intelligent, and portable for 21st-century India.

Gantt Chart



Thank You

